

ANDREW S. LIU

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EDUCATION

PhD in Astronomy and Astrophysics
The University of Chicago, Chicago, Illinois

Sept 2026 –

AB in Physics
Princeton University, Princeton, New Jersey
Magna Cum Laude *Physics GPA: 3.93/4.0*

Sept 2021 – May 2025

Visiting Student in Physics
The University of Oxford, St. Edmund Hall, Oxford, England

Jan 2024 – Jun 2024

RESEARCH EXPERIENCE

SCIPP Theory Group
University of California, Santa Cruz

November 2024 – November 2025
Santa Cruz, California

- Characterized effects of Long-Ranged Self-Interacting Dark Matter model on early structure formation
- Developed a magnetohydrodynamic model to include gravity and a modified Ohm's law to accurately describe the evolution of a dark sector plasma
- Utilized Python to numerically solve for the alterations to the linear Matter Power Spectrum and Transfer Function under this model
- Advised by Professor Stefano Profumo

Princeton Particle Phenomenology Group
Princeton University

May 2024 – Present
Princeton, New Jersey

- Characterized effects of Long-Ranged Self-Interacting Dark Matter, in the case where the interaction is mediated by a dark sector vector $U(1)$ gauge boson, on a dark matter subhalo streaming through its host
- Ran Particle-In-Cell simulations to study dark- $U(1)$ plasma instabilities in the nonlinear regime and characterize subhalo evaporation under this model
- Utilized the Python Package *galpy* to understand the subhalo mass loss due to plasma effects along orbits around a host halo
- Advised by Professor Mariangela Lisanti

Spring Junior Paper Research
The University of Oxford

Jan 2024 – Apr 2024
Oxford, England

- Studied Fast Radio Bursts (FRBs) as probes of missing baryonic matter in Λ CDM cosmology
- Proposed improved methodology for computing Dispersion Measure (DM), including local-environment contributions
- Outlined MCMC strategies for analyzing datasets from localized FRBs
- Advised by Professor Aris Karastergiou

Fall Junior Paper Research
Princeton University

Sept 2023 – Jan 2024
Princeton, New Jersey

- Looked at wavefunctions and production mechanisms of open-charmed tetraquarks
- Proposed novel production mechanisms from B meson and bottom baryon decays
- Developed wavefunction models for observed tetraquark states
- Advised by Professor Igor Klebanov

LPSC Theory Group Intern

Laboratory of Subatomic Physics and Cosmology (LPSC)

Jun 2023 – Aug 2023

Grenoble, France

- Studied Hawking evaporation of primordial black holes (PBHs) and effects on classical systems
- Investigated spacetime excitations from graviton and gravitational-wave emission
- Analyzed PBH-dominated early-universe cosmology and classified radiation/matter epochs

William Jones Group Undergraduate Research Intern

Princeton University / NASA Columbia Scientific Balloon Facility

Jun 2022 – Aug 2022

Princeton, NJ & Palestine, TX

- Supported preparation for SPIDER, a balloon-borne CMB polarization telescope
- Worked at NASA's Columbia Scientific Ballooning Facility
- Assembled and tested sun shields, gyroscope, and accelerometer components

PUBLICATIONS

Andrew Liu, Ani Prabhu, Akaxia Cruz, & Mariangela Lisanti. *Non-linear Evolution of Dark Plasma Subhalos*. **arXiv:2607.27312** (2026).

Pierce Giffin, **Andrew Liu**, Jeremias Boucsein, Akaxia Cruz, Ani Prabhu, Stefano Profumo, & M. Grant Roberts. *Structure Formation with Dark Magnetohydrodynamics*. **Phys. Rev. D** **113**, 063063 (2026).

Aurélien Barrau, Baptiste Blachier, Maxime Lahlou, **Andrew Liu**, & Killian Martineau (2023). *Einstein vs Hawking: black hole binaries and cosmological expansion*. **Eur. Phys. J. C** **83**, 1025.

TALKS

Andrew Liu, "Galactic Probes of Long-Range Self-Interacting Dark Matter." Galaxy Formation and Evolution in Southern California, 17 Sept. 2025, University of California, San Diego

WORK EXPERIENCE

Undergraduate Peer Tutor

Princeton University Department of Physics

Jan 2023 – May 2025

Princeton, New Jersey

- Provided one-on-one tutoring on physics concepts and homework
- Courses tutored: PHY 105: Advanced Physics - Mechanics, PHY 106: Advanced Physics - Electromagnetism, PHY 207: From Classical to Quantum Mechanics, PHY 208: Principles of Quantum Mechanics, PHY 301: Thermal Physics, PHY 305: Introduction to Quantum Theory

Undergraduate Course Assistant

Princeton University Department of Mathematics

Jan 2023 – May 2023

Princeton, New Jersey

- Tutored MAT 215: Single Variable Analysis with an Introduction to Proofs
- Held weekly office hours

OTHER EXPERIENCES

Events Committee & Undergraduate EDI Working Group Leader

Princeton Physics EDI Initiative

Sept 2023 – May 2025

Princeton, New Jersey

- Leader of Undergraduate Matters EDI committee
- Implement initiatives supporting diversity and well-being within the Physics Department
- Helped outline a 5-year reform plan focused on undergraduate issues
- Organized EDI events and guest speakers for diversity panels

AWARDS AND HONORS

Manfred Pyka Memorial Physics Prize (2021–2022)

Manfred Pyka Memorial Physics Prize (2022–2023)

Allen G. Shenstone Prize in Physics (2023–2024)

Allen G. Shenstone Prize in Physics (2024–2025)